

Three Approaches for Multi-Peak Orientation Estimation

Input Point Clouds

(5 Symmetry Types)

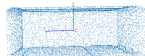
1-Front
e.g., Chair



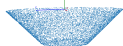
2-Front
e.g., Bookshelf



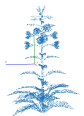
4-Front
e.g., Glass Box



Rot-Sym
e.g., Bowl

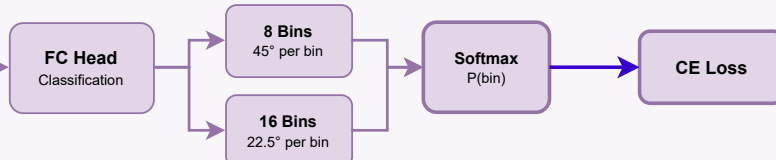


No-Front
e.g., Flower

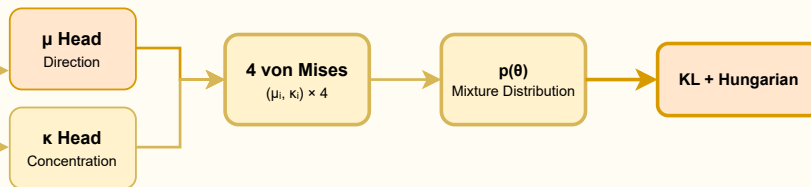


**PointNet++
Backbone**
(Shared)

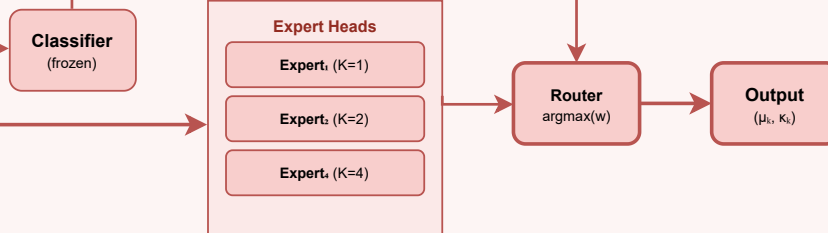
Method 1: Discrete Classification



Method 2: Mixture of von Mises (MvM)



Method 3: Mixture of Experts (MoE)



Discrete MvM MoE Shared Backbone